Dezyne E'cole College
Business • Design • Computers

NEXORA

SMART CITY SELF-HEALING SYSTEM

RESILIENCE REPORT

Monitoring • Detection • Recovery • Data Healing

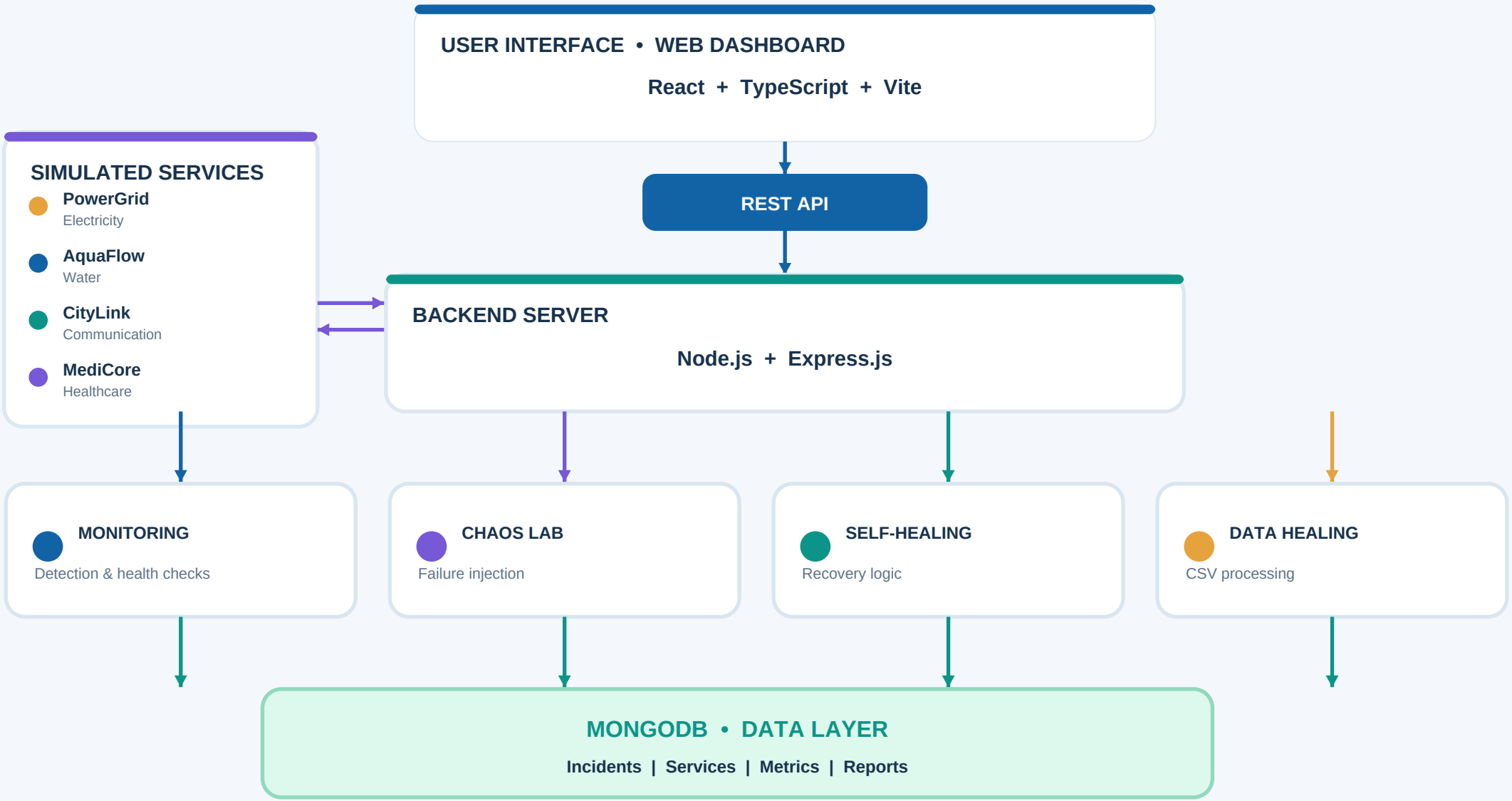
COURSE	**BCA — 5th Semester**
PROJECT	NEXORA Smart City Self-Healing System
TEAM	**Preeti Chablani • Bhavika Sewani**

A smarter approach to resilient services, automated recovery and reliable data.

2026

02 System Architecture

A modular frontend–backend design with simulated services and persistent incident records



Flow: Dashboard → REST API → Backend → system modules → MongoDB

03 Failure Scenarios & Recovery Workflow

Detection, remediation and health verification for predefined failure types

#	FAILURE TYPE	DETECTION	RECOVERY ACTION	RECOVERY TIME
1	**PROCESS_CRASH**	Worker health check	Supervisor restarts worker	8–12 sec
2	**DEPENDENCY_TIMEOUT**	Dependency response timeout	Retry with backoff + verify	12–20 sec
3	**INVALID_RECORD**	Validation flags invalid data	Quarantine + restore safe fixture	10–18 sec
4	**REQUEST_OVERLOAD**	Rate limit / high request threshold	Limit requests + drain workload	15–25 sec
5	**UNOBSERVABLE_FAULT**	No monitoring signal detected	No automatic recovery	N/A

SELF-HEALING WORKFLOW



Note: recovery times are reported ranges; UNOBSERVABLE_FAULT has no automatic recovery (N/A).

04 Data Healing & Before / After Analysis

CSV validation, repair and review pipeline



BEFORE HEALING

Missing values	42
Invalid values	18
Duplicate records	7
Malformed rows	3

AFTER HEALING

Values repaired	53
Records repaired	38
Duplicates removed	7
Records needing review	2

INPUT RECORDS	OUTPUT RECORDS	REPAIRED / FIXED	REVIEW REQUIRED
1,000	952	91	2

05 Results, Limitations & Contributions

Latest run summary and project notes

DETECTION RATE	RECOVERY RATE	AVG. RECOVERY TIME	FAILED OUTCOMES
100%	80%	14.5 sec	1
5 / 5	4 / 5	approx.	UNOBSERVABLE_FAULT

SYSTEM AREAS

01 Dashboard Overview
System health and key metrics

02 Chaos Lab
Failure injection and testing

03 Data Healing Report
CSV quality and repair results

04 Incident History
Recorded incidents and outcomes

KNOWN LIMITATIONS

- Services are simulated, not real infrastructure.
- Recovery is limited to predefined scenarios.
- UNOBSERVABLE_FAULT may not be detected or recovered automatically.
- CSV repair depends on supported columns and rules.
- Some records need manual review; recovery times vary by run.

CHATGPT CONTRIBUTION & TEAM

Support described in the original report:

- Debugging explanations and error fixes
- Code suggestions, structure and documentation
- Architecture and workflow design guidance

Final implementation and testing completed by the team.

TEAM **Preeti Chablani • Bhavika Sewani**
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